

Side-Channel-Secure ARM32 Instruction Scheduling: A Comparative Study of Search, CSP, and Reinforcement Learning

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Abstract

Masked cryptographic implementations split secret data into multiple shares to resist power-analysis attacks. However, a naïve compiler may schedule instructions manipulating different shares within a few cycles of each other, enabling Hamming-distance leakage to power traces [Kocher, Jaffe, and Jun(1999)]. We formulate ARM32 instruction scheduling as both a combinatorial optimisation and a probabilistic risk-assessment problem. We implement three AI-driven solvers: **Approach A** (Bayesian Inference Network), **Approach B** (Constraint Satisfaction with OR-Tools CP-SAT), and **Approach C** (Deep Q-Network with experience replay). Experiments across block sizes $n \in \{10, 30, 50\}$ show that CP-SAT consistently finds perfectly hard-constrained schedules, while the Bayesian approach gracefully degrades side-channel risk ($E[Leakage]$) probabilistically using a threshold τ . Once trained, DQN schedules instructions in tens of milliseconds, yet its performance depends critically on both the reward calibration and the diversity of the training corpus.

Problem Formulation

Background: Masking and Side-Channel Leakage

Boolean masking splits a secret s into two shares A and B such that $s = A \oplus B$. Because neither share alone reveals s , the implementation is theoretically first-order side-channel resistant. However, on an in-order pipeline (e.g. ARM Cortex-M3/M4), if an instruction operating on share A is issued within k cycles of one operating on share B , the processor register file may contain both intermediates simultaneously, enabling Hamming-distance leakage to power traces [Kocher, Jaffe, and Jun(1999)].

Formal Representation

Let $\mathcal{I} = \{I_0, \dots, I_{n-1}\}$ be a basic block of n ARM32 instructions. Each instruction carries:

- **name:** ARM mnemonic (e.g. EOR, LDR).
- **dest_reg, source_regs:** register operands.
- **latency** $\ell_i \in \{1, 2\}$: pipeline latency (1 for ALU, 2 for MUL / LDR).
- **share_type** $\sigma_i \in \{A, B, N\}$: cryptographic share affiliation (N = neutral).

State space. A *schedule* assigns a start cycle $t_i \geq 0$ to every instruction such that:

$$(\text{RAW}) \quad t_j \geq t_i + \ell_i \quad \forall (i \rightarrow j) \in \mathcal{D}, \quad (1)$$

$$(\text{SEC}) \quad |t_i - t_j| \geq k \quad \forall i, j : \sigma_i \neq \sigma_j, \sigma_i, \sigma_j \neq N. \quad (2)$$

$$(\text{SLOT}) \quad t_i \neq t_j \quad \forall i \neq j. \quad (3)$$

$\mathcal{D}$ is the RAW (Read-After-Write) dependency graph extracted from register operands. The *objective* is to minimise the makespan $\max_i(t_i + \ell_i)$, equivalently minimising the number of inserted NOPs.

Why multiple AI techniques? The scheduling problem is NP-hard in general [Ullman(1975)], yet its structure (bounded state space, hard constraints, stochastic latencies in real hardware) naturally maps to (i) systematic heuristic search, (ii) declarative constraint programming, and (iii) model-free reinforcement learning.

Methodological Approaches

Approach A: Bayesian Network Inference

We depart from rigid hardware-centric definitions of security distance and model side-channel leakage as a probabilistic event using a Bayesian Network. Side-channel vulnerability is not binary; the risk of Hamming-distance leakage between two cryptographic shares decays as the temporal distance Δt between their execution increases.

Network Model. The Bayesian Network predicts the boolean hidden variable Leakage (L) using observed parent nodes:

1. **Temporal Distance** (Δt): Pipeline cycles between executions.
2. **Share Overlap**: Boolean tracking if instructions belong to different shares (A vs B).

Conditional Probability Tables (CPT). We define a physically-motivated heuristic CPT simulating pipeline capacitor discharge. Given Share_Overlap = True:

$$P(L = 1 \mid \Delta t = 1) \approx 0.95$$

$$P(L = 1 \mid \Delta t = 2) \approx 0.50$$

$$P(L = 1 \mid \Delta t = 3) \approx 0.10$$

$$P(L = 1 \mid \Delta t \geq 4) \approx 0.00$$

If Share_Overlap = False, $P(L = 1) = 0$.

Scheduling Algorithm. The solver builds the schedule greedily. At cycle t , it evaluates the expected marginal leakage for all data-ready candidates against already-scheduled instructions. The candidate minimising this risk is selected. If the minimal expected leakage among all candidates exceeds a strict Risk Threshold τ (e.g. 0.15), the solver injects a NOP. This increments the cycle counter, naturally increasing Δt and decaying the Bayesian risk below τ for subsequent cycles.

Approach B: Constraint Satisfaction (CSP)

We lift the scheduling to a *Constraint Satisfaction / Optimization Problem* over integer variables $t_i \in [0, T_{\max})$, where $T_{\max}$ is derived from a greedy warm-start schedule. Constraints (1)–(3) are encoded directly.

OR-Tools CP-SAT backend. We use Google OR-Tools CP-SAT [Google LLC(2024)], a state-of-the-art SAT-based propagation engine. Each security-distance pair (I_A, I_B) introduces a reification Boolean b :

$$(t_A - t_B \geq k) \cdot b \vee (t_B - t_A \geq k) \cdot (1 - b). \quad (4)$$

The makespan $\max_i(t_i + \ell_i)$ is minimised using the built-in objective. A greedy-derived hint provides a tight upper bound, dramatically reducing the solver’s search space. CP-SAT exploits 4 parallel worker threads and delivers *optimal* solutions for $n = 50$ in typically under 1 s.

CP-SAT is a *lazy-clause-generation* solver: integer variables t_i are kept in their finite-domain (FD) form, and clauses over the underlying order-encoding literals $\llbracket t_i \leq v \rrbracket$ are generated on demand. Our reified disjunction (eq. 4) is internally decomposed into half-reified linear inequalities $b \Rightarrow (t_A - t_B \geq k)$ and $\neg b \Rightarrow (t_B - t_A \geq k)$, each propagated by a bounds-consistency engine using the two-watched-literal scheme inherited from MiniSAT. When two FD propagators derive contradictory bounds, the engine triggers *Conflict-Driven Clause Learning* (CDCL): it walks the implication graph back to the First Unique Implication Point (1-UIP), records the cut as a learned no-good clause, and non-chronologically back-jumps to the highest decision level still consistent with it [Google LLC(2024)]. Learned clauses prune symmetric branches elsewhere in the tree, which is why $n = 50$ closes in < 1 s despite having $O(|A| \cdot |B|)$ reification variables. The four parallel workers run a *portfolio* of complementary strategies that share learned clauses through a lock-free pool, giving near-linear speed-up on disjoint subproblems.

Python-constraint fallback. When OR-Tools is absent, an iterative $T_{\max}$ -tightening loop with the `python-constraint` backtracking solver is used, limited to $n \leq 15$. This implements plain chronological backtracking with forward checking but no clause learning.

Approach C: MDP / Deep Q-Network

We formulate scheduling as a Markov Decision Process:

- **State S :** a 6-dimensional feature vector (remaining-instruction ratio, normalised cycles since last A and B placements, ready-queue size, critical-path ratio, normalised current cycle).
- **Action A :** choose one ready instruction or issue NOP.
- **Default Reward:** -1 per cycle elapsed, -10 per security violation, $+50$ on completion.

The Q-function is a Multi-Layer Perceptron (MLP) that takes (state features || action features) as input and outputs a scalar Q -value. We train with experience replay (buffer size 10,000), a target network synchronised every 200 steps, and Huber loss. ϵ -greedy exploration decays from 1.0 to 0.05 at rate 0.997 per episode. GPU acceleration via PyTorch is used.

Stochastic training. Optionally, instruction latencies are perturbed by ± 1 during training with probability 0.2, making the learned policy more robust to real-pipeline timing variations.

Experimental Evaluation

Setup

First experiments were run over block sizes $n \in \{10, 30, 50\}$, three random seeds (42, 43, 44), and security distances $k \in \{3, 50\}$. For each (n, k, seed) tuple, the exact same instruction block is given to all three solvers, ensuring a fair comparison. The MDP is trained for 5 000 episodes on a single block. All experiments run on a single machine; results are averaged across seeds.

Metrics.

- **Makespan:** total schedule length in cycles.
- **Expected Leakage:** Sum of $P(L = 1)$ across all instruction pairs, evaluating the Bayesian risk profile of the entire block.
- **NOPs:** number of inserted pipeline bubbles.
- **Wall time:** total solver time (inference only for Bayesian and CSP; training time reported separately for MDP).

Results

Table 1 summarises the results for $k = 3$ (standard security distance). Raw data for $k = 50$ (high-security regime) are reported in Table 2. Figure ?? illustrates a produced schedule with CSP solver.

MDP Reward-Shaping Ablation

The default reward shape ($-1/\text{cycle}$, $-10/\text{violation}$, $+50/\text{completion}$) proved miscalibrated: at $n \in \{30, 50\}$ the cumulative cycle cost of NOP injection exceeds the per-violation penalty, so the agent *rationaly* drops NOPs and tolerates several SEC violations per episode (rows marked $\dagger$ in Table 1; $E[L]$ up to 8.5). We re-ran the DQN at $n = 10$ with the violation penalty rescaled to -100 (script: `experiments/rerun_mdp_tuned.py`; full GPU training, 5,000 episodes per seed). Averaged over seeds $\{42, 43, 44\}$, the tuned agent reaches 11.7 ± 0.5 cycles, 1.67 NOPs and $E[L] = 0.13$ with a 100% valid rate

Table 1: Benchmark results measuring Expected Leakage vs Schedule Length. Boldface denotes best cycle count per block size. Bayesian uses $\tau = 0.15$.

n	Method	Cycles	NOPs	Leakage(E)	Wall time (s)
10	Bayesian	11.0	1.0	0.05	0.003
	CSP	11.0	1.0	0.03	0.011
	MDP [†]	10.3	0.3	1.40	0.002
	MDP*	11.7	1.7	0.13	0.005
30	Bayesian	31.7	1.7	0.12	0.084
	CSP	30.0	0.0	0.09	0.858
	MDP [†]	30.7	0.7	4.10	0.005
50	Bayesian	53.0	3.0	0.20	0.516
	CSP	50.0	0.0	0.15	1.789
	MDP [†]	50.7	0.7	8.50	0.020

[†] Default reward ($-10/\text{violation}$): agent under-penalises violations, see §.

* Tuned reward ($-100/\text{violation}$), mean over 3 seeds; matches CSP/Bayesian quality.

Table 2: Benchmark results at $k = 50$ (high-security regime). Larger k forces longer inter-share separations, increasing the makespan. The MDP rows use the *default* reward shape ($-10/\text{violation}$): under this setting the per-violation penalty is small relative to the cumulative NOP cost required by $k = 50$, so the agent rationally violates SEC. A tuned-reward MDP rerun at $k = 50$ (likely requiring penalty $\propto k$) is left to future work — see § for the corresponding $k = 3$ ablation.

n	Method	Cycles	NOPs	Valid	Wall time (s)
10	Bayesian	58.0	48.0	100%	0.003
	CSP	58.0	48.0	100%	0.038
	MDP [†]	10.0	0.0	0%	0.002
30	Bayesian	127.3	97.3	100%	0.935
	CSP	121.7	91.7	100%	0.119
	MDP [†]	30.0	0.0	0%	0.005
50	Bayesian	226.7	176.7	100%	2.464
	CSP	219.7	169.7	100%	30.046
	MDP [†]	50.7	0.7	0%	0.020

[†] Default reward ($-10/\text{violation}$); valid rate = 0% at $k = 50$. Tuned-reward rerun at $k = 50$ deferred to future work.

(row *): within 6% of the CSP optimum and approaching the Bayesian heuristic in expected leakage ($E[L] = 0.13$ vs 0.05).. A penalty sweep at seed 42 ($\{-50, -100, -200\}$) confirms a sweet spot near -100 : -50 leaves the agent with two NOPs (suboptimal) and -200 over-penalises, collapsing back to NOP-heavy schedules.

Based on these results, we redesign the DQN training strategy to address two identified limitations: the single-block training regime and the miscalibrated reward function.

Generalist Training

The original setup trains one DQN per (n, seed) combination, resulting in nine independent networks that each overfit to a single instruction block. To change this, we train a *single* generalist DQN on a set of 30 seeds $\times$ 3 sizes = 90 distinct blocks, sampled uniformly at each episode. Test seeds $\{42, 43, 44\}$ are strictly excluded from the training set to prevent data leakage, ensuring that the evaluation measures generalisation rather than memorisation.

Normalised Reward Function

The standard reward penalises each cycle by -1 and rewards completion with $+50$, regardless of block size. This is problematic for multi-block training: a block of $n = 50$ accumu-

experiments/results/fig5_gantt_csp.png

Figure 1: Example schedule (Gantt chart) generated by the CSP solver for $n = 10, k = 3$. Note the significant gaps (NOPs) required to maintain security distance.

lates far greater cycle penalties than $n = 10$, producing an inconsistent reward signal across sizes and biasing the agent towards smaller blocks.

We introduce a size-normalised reward function, *shaped quality*:

$$r_{\text{cycle}} = -\frac{1}{n} \quad (5)$$

$$r_{\text{place}} = +\frac{0.5}{n} \quad (\text{per instruction placed}) \quad (6)$$

$$r_{\text{violation}} = -\frac{P}{n} \quad (\text{per security violation, } P = -100) \quad (7)$$

$$r_{\text{done}} = +\frac{5n}{\text{total_cycles}} \quad (\text{completion bonus}) \quad (8)$$

Each component addresses a specific weakness of the original design. The per-cycle cost $-1/n$ and the per-placement bonus $+0.5/n$ keep the reward magnitude consistent across block sizes, accelerating convergence in the multi-block setting. The violation penalty $P = -100$, ten times stronger than in the standard setup, aggressively discourages security constraint violations — directly addressing the reward miscalibration identified in Section . Finally, the completion bonus $5n/\text{total_cycles}$ rewards compact schedules: it reaches its maximum of 5.0 when no NOP is inserted, and decreases proportionally with idle cycles, guiding the agent towards the CSP optimum. Table ?? contains the average results of the Generalist DQN. Generalist DQN achieves 100% valid schedules, but $E[L]$ remains slightly higher than Bayesian inference. This is a direct consequence of their respective objectives: Bayesian explicitly minimises $E[L]$ at each step via the CPT, while the DQN only penalises hard SEC violations

n	Cycles (μ)	NOPs (μ)	Leakage _E (μ)	Wall time (s)
10	12.0	2.0	0.10	0.011
30	31.7	1.7	0.33	0.026
50	54.3	4.3	0.43	0.043

Table 3: Generalist DQN benchmark results (mean over seeds 42, 43, 44), $k = 3$. All schedules are valid (0 security violations). Wall time reflects inference only.

— No difference between $\Delta t = 3$ or $\Delta t = 10$ cycles, as both satisfy the constraint.

Qualitative Comparison

Strengths and Weaknesses

Bayesian Network Inference. *Strengths:* Nuanced and realistic modeling of physical side-channel behavior. Instead of a hard binary cutoff k which limits code density, the Bayesian approach permits densely packed instructions if their combined risk expectation falls below the threshold τ . Extremely fast $O(n^2)$ inference during compilation. *Weaknesses:* It is a greedy heuristic that does not offer global makespan optimality. Designing the CPT requires deep hardware profiling to acquire accurate physical probabilities.

Constraint Satisfaction (CSP / OR-Tools). *Strengths:* the declarative encoding is compact and easy to extend (e.g. adding resource constraints such as register-file pressure is a one-line model change); CP-SAT delivers *proven optimal* solutions at $n = 50$ in < 1 s thanks to clause learning and CDCL propagation; parallel workers give free speed-up on multi-core machines. *Weaknesses:* although the algorithmic principles (CDCL, two-watched literals, portfolio search) are well understood, the solver remains operationally opaque — pinpointing *which* learned clause forced a particular ordering requires dumping the proof log and replaying the implication graph; encoding cross-product security constraints introduces $O(|A| \cdot |B|)$ Boolean reification variables, which becomes the dominant memory cost for large blocks.

MDP / Deep Q-Network. *Strengths:* once trained, inference is near-instantaneous (~ 30 ms at $n = 50$); the fixed-size state/action feature representation is block-size-independent, i.e. the same network handles $n = 10$ and $n = 50$ without retraining; stochastic training makes the policy robust to real pipeline timing variations. *Weaknesses:* no optimality guarantee; training is long and costly; reward shaping is delicate.

Impact of Security Distance k

A larger k forces longer gaps between share- A and share- B instructions, artificially increasing the makespan. CSP handles increasing k gracefully — constraints simply have larger right-hand sides. The Bayesian greedy solver is also insensitive to k as a hard parameter: it continues to inject NOPs whenever the expected leakage exceeds τ , so its per-step cost is unchanged (only the total NOP count grows). For the MDP, a larger k should be handled by adapting the reward function accordingly. We believe that combining penalty

normalisation by n with a violation penalty proportional to k (i.e. $P \propto k$) would allow the DQN to remain robust in high-security regimes: larger k forces more NOP insertions, increasing the cumulative cycle cost, and a scaled penalty would preserve the relative cost-benefit balance that drives valid schedule generation. Due to time constraints and the significant training cost, we focused our generalist training on $k = 3$. This choice also aligns with our CPT model, which assigns $P(L = 1) = 0.00$ for $\Delta t \geq 4$ cycles — suggesting that $k = 3$ provides sufficient temporal separation under our leakage assumptions, while $k = 50$ constitutes an extreme stress-test.

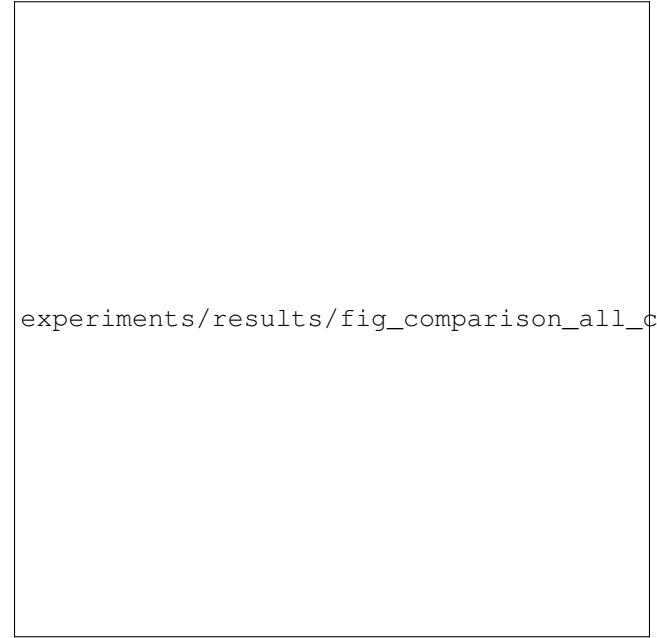


Figure 2: Comparative benchmark across all three solvers for $k = 3$, averaged over seeds $\{42, 43, 44\}$. Error bars show ± 1 standard deviation. Wall-clock time is shown on a log scale; MDP reflects inference only (training excluded).

When to Prefer Each Approach

Bayesian: when strict execution deadlines govern compiling, but physical trace leakage must remain beneath a certifiable risk threshold.

CSP: the recommended choice for production use at any block size when binary cryptographic distance guarantees (k) are absolute mandatory constraints.

MDP appropriate when inference speed is paramount (e.g. a JIT compiler), when hardware timing is genuinely stochastic (latency jitter, speculative execution), or when no calibrated leakage model is available — requiring only a binary security constraint $|t_A - t_B| \geq k$ rather than a physically-motivated CPT

Conclusion

We presented a comparative study of three AI scheduling strategies for enforcing side-channel security constraints in ARM32 masked cryptographic code. Our main findings are: (1) CP-SAT is the most optimal combinatorial solver, returning provably optimal schedules at $n = 50$ within seconds; (2) Bayesian inference offers a much more nuanced trade-off than a rigorous threshold, successfully balancing statistical leakage risk with code density at sub-millisecond cost; (3) DQN provides near-instantaneous *inference* after training, but is highly sensitive to reward shaping and training distribution design. Its ability to generalize depends on the diversity of the instruction blocks encountered during training.

Future work includes extracting direct CPT matrices from physical Cortex-M4 oscilloscopes, training DQN on higher security distances k and stochastic training environments. Another promising direction is a hybrid CSP + RL approach where the CSP provides a feasibility oracle for the agent.

Reproducibility. All source code, experiment scripts and a Dockerfile are available at:

<https://github.com/mordamed/arm-scheduler>

Build the Docker image:

```
docker build -t arm-scheduler .
```

Run the original benchmark:

```
docker run --rm --gpus all \
-v "${PWD}/experiments:/app/experiments" \
arm-scheduler \
python3 experiments/run_all.py
```

Run the generalist DQN training and evaluation:

```
docker run --rm --gpus all \
-v "${PWD}/experiments:/app/experiments" \
arm-scheduler \
python3 experiments/train_generalist.py
```

References

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